

CINCINNATI LUNKEN, OH, USA

WMO: 724297

Lat: 39.103N

Lon: 84.419W

Elev: 149

StdP: 99.54

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 93812

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-10.8	-19.5	0.7	-12.7	-16.7	0.9	-9.6	10.8	6.2	9.4	3.4	2.8	290	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	33.5	23.9	32.1	23.5	30.9	23.0	25.7	30.9	25.0	29.9	24.3	28.8	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.2	19.5	28.2	23.5	18.7	27.4	22.9	17.9	26.7	79.8	30.9	76.5	30.0	73.6	28.8	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.1	8.0	7.1	-17.6	35.4	4.4	1.9	-20.8	36.7	-23.4	37.8	-25.9	38.8	-29.1	40.2		
			-18.1	27.3	4.2	0.9	-21.2	27.9	-23.7	28.5	-26.0	29.0	-29.1	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	12.8	0.0	1.9	6.7	12.7	17.9	22.4	24.3	23.8	20.1	13.5	7.0	2.5
	DBStd	9.79	6.57	6.05	5.65	4.73	4.16	3.08	2.70	2.65	3.77	4.62	4.95	5.39
	HDD10.0	1088	318	233	136	26	2	0	0	0	0	17	116	241
	HDD18.3	2653	569	459	363	176	61	6	0	1	25	162	339	492
	CDD10.0	2106	7	7	32	107	246	373	444	428	302	126	27	7
	CDD18.3	631	0	0	1	7	47	129	187	171	77	12	0	0
	CDH23.3	5806	0	1	16	96	445	1157	1732	1521	706	130	3	0
	CDH26.7	2106	0	0	1	11	117	410	693	591	254	29	0	0
Wind	WSAvg	2.7	3.5	3.3	3.3	3.3	2.6	2.3	2.1	1.9	1.9	2.3	2.7	2.9
Precipitation	PrecAvg	1045	66	62	98	93	113	91	102	98	85	71	83	78
	PrecMax	1398	186	166	281	161	260	187	200	168	216	215	178	204
	PrecMin	728	12	8	32	17	16	20	15	18	13	15	16	11
	PrecStd	159	35	39	50	40	62	39	53	42	57	45	40	39
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.8	21.0	25.7	28.1	31.2	33.6	35.8	35.4	34.3	29.7	23.1	19.1
		MCWB	15.2	15.4	17.1	18.8	22.2	24.1	25.7	23.8	21.6	20.8	16.7	15.7
	2%	DB	16.0	17.3	22.4	26.1	29.4	32.1	33.5	33.1	31.8	26.6	20.4	15.9
		MCWB	13.6	12.5	15.0	17.8	21.0	23.6	24.7	23.6	22.2	19.3	15.1	13.1
	5%	DB	13.2	14.5	19.1	24.0	28.0	30.7	32.0	31.5	29.8	24.3	18.1	13.5
		MCWB	10.8	10.9	13.2	16.5	20.7	22.9	23.9	23.4	21.4	18.0	14.0	11.2
	10%	DB	9.9	11.6	16.4	21.8	26.1	29.2	30.5	30.0	27.5	22.0	15.8	11.0
		MCWB	7.4	8.2	12.0	15.4	19.7	22.2	23.4	22.6	20.5	17.0	12.1	8.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	16.5	18.2	20.6	23.9	25.9	26.9	26.3	25.0	23.0	18.5	16.4
		MCDB	17.8	20.2	24.4	26.4	28.9	31.4	32.6	31.4	30.0	27.7	21.7	18.8
	2%	WB	14.0	13.9	16.3	18.9	22.7	24.9	25.9	25.4	23.9	20.9	16.4	13.8
		MCDB	15.9	16.2	20.7	23.8	27.6	30.0	31.3	30.5	28.6	24.4	19.3	15.4
	5%	WB	11.1	11.3	14.6	17.6	21.6	24.0	25.2	24.7	22.9	19.3	14.6	11.5
		MCDB	12.9	13.8	18.1	21.9	26.3	29.0	30.2	29.2	27.2	23.0	17.5	13.1
	10%	WB	7.7	8.1	12.4	16.4	20.5	23.2	24.4	23.9	21.8	17.7	12.6	8.8
		MCDB	9.5	11.3	16.0	20.8	24.5	27.8	28.8	28.0	25.5	20.7	15.4	10.6
Mean Daily Temperature Range	5% DB	MDBR	8.8	9.9	11.0	12.6	11.8	11.3	11.1	11.5	12.4	12.2	11.0	8.6
		MCDBR	11.6	14.8	15.5	15.7	14.0	13.1	12.5	13.0	15.1	15.4	14.4	12.1
		MCWBR	9.5	10.9	9.3	8.7	6.9	5.8	4.9	5.1	6.3	8.3	9.8	9.8
	5% WB	MCDBR	10.4	13.1	13.3	13.1	11.9	11.4	11.0	11.0	12.1	12.9	12.2	10.8
		MCWBR	9.1	10.8	8.9	8.0	6.4	5.6	4.9	5.0	5.8	7.5	9.3	9.6
Clear-Sky Solar Irradiance	taub	0.311	0.329	0.359	0.401	0.445	0.460	0.469	0.464	0.410	0.361	0.332	0.308	
	taud	2.476	2.388	2.372	2.261	2.176	2.180	2.164	2.162	2.290	2.431	2.472	2.493	
	Ebn at Noon	864	891	897	875	840	825	814	807	832	844	830	840	
	Edh at Noon	79	99	111	132	147	147	148	145	120	94	79	73	
All-Sky Solar Radiation	RadAvg	1.80	2.58	3.62	4.87	5.49	6.11	5.88	5.47	4.50	3.24	2.21	1.53	
	RadStd	0.17	0.33	0.32	0.38	0.48	0.35	0.35	0.30	0.45	0.33	0.26	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (34 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+117	+72

Nomenclature: See separate page